SimpleRA Phase 1 Test Cases

Total Marks : 200

Format

Input Command → Output file to be checked

Please note: The 1000000 graphs are called big in the feedback on Moodle.

LOAD GRAPH

5 test cases, 2 marks each → total = 5×2 = 10 marks

- LOAD GRAPH 10 D → check_load10.txt
- LOAD GRAPH 100 U → check_load100.txt
- LOAD GRAPH 500 D → check_load500.txt
- LOAD GRAPH 10000 U → check_load10000.txt
- LOAD GRAPH 1000000 D → check_loadbig.txt

EXPORT GRAPH

5 test cases, 2 marks each → total = 5×2 = 10 marks

NOTE: EXPORT commands do not have an output file, the checking will need to be done manually by checking the /data folder to see if the file has been created/modified.

- EXPORT GRAPH 10
- EXPORT GRAPH 100
- EXPORT GRAPH 500
- EXPORT GRAPH 10000
- EXPORT GRAPH 1000000

PRINT GRAPH

4 test cases, 2 marks each → total = 4×2 = 8 marks

- PRINT GRAPH 10 → check_print10.txt
- PRINT GRAPH 100 → check_print100.txt
- PRINT GRAPH 500 → check_print500.txt
- PRINT GRAPH 10000 → check_print10000.txt

PATH

5 test cases, 4 marks each → total = 5×4 = 20 marks

- RES_10 ← PATH 10 1 10 WHERE A1(N) == 1 → check_path10.txt
- RES_100 ← PATH 100 1 100 WHERE A49(N) == 1 AND ANY(E) → check_path100.txt
- RES_G_500_D ← PATH 500 19 467 WHERE A1(N) == 1 AND B1(E) == 0 AND A2(N) == 0 AND A1(N) == 1 AND B2(E) == 0 AND A3(N) == 0 AND B3(E) == 1 AND A4(N) == 1 AND A1(N) → check_path500.txt
- RES_G_10000_U ← PATH 10000 1 407 WHERE B2(E) → check_path10000.txt
- RES_1000000 ← PATH 1000000 450939 1000000 WHERE ANY(E) == 1 → check_pathbig.txt

DEGREE

5 test cases, 2 marks each → total = 5×2 = 10 marks

- DEGREE 10 6 → check_degree10.txt
- DEGREE 100 67 → check_degree100.txt
- DEGREE 500 248 → check_degree500.txt
- DEGREE 10000 9510 → check_degree10000.txt
- DEGREE 1000000 252414 → check_degreebig.txt

Mixed Test Case 1 - 13 marks

This has binary marking, full marks are only awarded if all the following test cases pass when run sequentially. 6.5 has been given for minor formatting

mistakes. There are 4 commands in this test case to be run in succession.

- LOAD GRAPH 500 D
- Res ← PATH 500 19 467 WHERE A1(N) == 1 AND B1(E) == 0 AND A2(N) == 0 AND A1(N) == 1 AND B2(E) == 0 AND A3(N) == 0 AND B3(E) == 1 AND A4(N) == 1 AND A1(N)
- Res ← PATH 500 1 3
- PRINT GRAPH Res

Final output file: check_print500_mixed.txt

Mixed Test Case 2 - 14 marks

This has binary marking, full marks are only awarded if all the following test cases pass when run sequentially. 7 has been given for minor formatting mistakes. There are 4 commands in this test case to be run in succession.

- LOAD GRAPH 100 U
- Res ← PATH 100 1 100 WHERE A49(N) == 1 AND B201(E) == 0
- PRINT GRAPH Res
- DEGREE Res 67

Final output file: check_degree_res.txt

Mixed Test Case 3 - 13 marks

This has binary marking, full marks are only awarded if all the following test cases pass when run sequentially. 6.5 has been given for minor formatting mistakes. There are 4 commands in this test case to be run in succession.

- LOAD GRAPH 10 D
- LOAD 10
- PRINT 10
- LOAD GRAPH 10 D

Final output file: check_errorexistingload.txt

Error Handling Test Cases

6 test cases, 2 marks each → total = 6×2 = 12 marks

1. LOAD GRAPH DataSystems U → error_load.txt
2. PRINT GRAPH DataSystems → error_print.txt
3. EXPORT GRAPH DS → error_export.txt
4. LOAD GRAPH 10 D → check_load10.txt
5. Res ← PATH 10 100 1 → error_path.txt
6. DEGREE Res 12 → error_degree.txt

Report - 15 Marks

Block Accesses - 50 Marks

Viva - 25 Marks